Summary of Multiple LR

The sample: $\{(x_{1i}, x_{2i}, \ldots, x_{ki}, y_i)\}_{i=1}^{n}$

$$Y = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k}_{\text{True regression line}} + \varepsilon \qquad \varepsilon \sim N(0, \sigma^2)$$

- Significance of LR model
  - F-test for the whole model $H_0: \beta_i = 0$ for all i; $H_a: \beta_i \neq 0$ for at least one i
  - t-tests for individual independent variables. $H_0: \beta_i = 0$; $H_a: \beta_i \neq 0$
- Percentage of variability explained by the LR model
  - R^2 : no reference to df
  - Adjusted R^2 : adjusted according to df so as to minimize MSE
- Other criteria: C_p

1

Example

Let us now look at an example in which multiple regression is useful. A mobile ad hoc computer network consists of several computers (nodes) that move within a network area. Often messages are sent from one node to another. When the receiving node is out of range, the message must be sent to a nearby node, which then forwards it from node to node along a routing path toward its destination.

We wish to predict the proportion of messages that will be successfully delivered, which is called the goodput. It is known that the goodput is affected by the average node speed and by the length of time that the nodes pause at each destination.

2

Example

Data was collected on average node speed, average pause time, and goodput for 25 simulated mobile ad hoc networks.

Consider the following model:

$$\text{Goodput} = \beta_0 + \beta_1 \text{Speed} + \beta_2 \text{Pause} + \beta_3 \text{Speed} \cdot \text{Pause} + \beta_4 \text{Speed}^2 + \beta_5 \text{Pause}^2 + \varepsilon$$

3

Interpreting Output

Much of the output is analogous to that of simple linear regression.

1. The fitted regression equation is presented near the top of the output.
2. Below that the coefficient estimates and their estimated standard deviations.
3. Next to each standard deviation is the Student's *t* statistic for testing the null hypotheses that the true value of the coefficient is equal to 0.
4. The *P*-values for the tests are given in the next column.

4

Computer Output Part 1

The regression equation is

Goodput = 96.0 −1.82 Speed + 0.565 Pause
+ 0.0247 Speed* Pause + 0.0140Speed^2
− 0.0118 Pause^2

Predictor	Coef	SE Coef	T	P
Constant	96.024	3.946	24.34	0.000
Speed	−1.8245	0.2376	−7.68	0.000
Pause	0.5652	0.2256	2.51	0.022
Speed*Pa	0.024731	0.003249	7.61	0.000
Speed^2	0.014020	0.004745	2.95	0.008
Pause^2	−0.011793	0.003516	−3.35	0.003

$S = 2.942$ $R-Sq = 93.2\%$ $R-Sq(adj) = 91.4\%$

5

Analysis of Variance Table

5. The DF column gives the degrees of freedom, the degrees of freedom for regression is equal to the number of independent variables in the model. The degrees of freedom for “Residual Error” is the number of observations – number of parameters estimated. The total degrees of freedom is the sum of the degrees of freedom for regression and for error.
6. The next column is SS. This column gives the sum of squares, the first is regression sum of squares, SSR, the second is error sum of squares, SSE, and the third is the total sum of squares, SST = SSR + SSE.

6

More on the ANOVA Table

7. The column MS is the column with the mean sum of squares which is the sums of squares divided by their respective degrees of freedom. Note that the mean square error is equal to the variance estimate, s^2.
8. The column labeled *F* presents the mean square for regression divide by the mean square for error.
9. This is the *F* statistic that we discussed earlier that is used for testing the null hypothesis that none of the independent variables are related to the dependent variable.

7

Computer Output Part 2

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	5	2240.49	448.10	51.77	0.000
Residual Error	19	164.46	8.66		
Total	24	2404.95			

Predicted Values for New Observations

New Obs	Fit	SE Fit	95% CI	95% PI
1	74.272	1.175	(71.812, 76.732)	(67.641, 80.903)

Values of Predictors for New Observations

New Obs	Speed	Pause	Speed*Pause	Speed^2	Pause ^2
1	25.0	15.0	375	625	225

8

Using the Output

- From the output, we can use the fitted regression equation to predict for future observations.
- It is also possible to calculate residuals for a value of *y*.
- Constructing confidence interval for the coefficient of the independent variables is also possible from the output.

9

Example

Use the MINITAB output on the previous slides to answer the following:

1. Use the multiple regression model to predict the goodput for a network with speed 12 m/s and pause time 25 s.
2. For the goodput data, find the residual for the point Speed = 20, Pause = 30.
3. Find a 95% confidence interval for the coefficient of Speed in the multiple regression model.
4. Test the null hypothesis that the coefficient of Pause is less than or equal to 0.3.

10

Example

Use the MINITAB output on the previous slides to answer the following:

1. Use the multiple regression model to predict the goodput for a network with speed 12 m/s and pause time 25 s. 90.336
2. For the goodput data, find the residual for the point Speed = 20, Pause = 30. Observed: 84.937, predicted: 86.350
   Residual: -1.413
3. Find a 95% confidence interval for the coefficient of Speed in the multiple regression model.

   Coefficient: -1.8245, std error: 0.2376, df=19, t19,0.025=2.093
   -1.8245+/-2.093* 0.2376=(-2.3218, -1.3272)

11

Example

4. Test the null hypothesis that the coefficient of Pause is less than or equal to 0.3.

Predictor	Coef	SE Coef	T	P
Constant	96.024	3.946	24.34	0.000
Speed	−1.8245	0.2376	−7.68	0.000
Pause	0.5652	0.2256	2.51	0.022
Speed*Pa	0.024731	0.003249	7.61	0.000
Speed^2	0.014020	0.004745	2.95	0.008
Pause^2	−0.011793	0.003516	−3.35	0.003

The estimated coefficient of Pause is $\hat{\beta}_2 = 0.5652$, with standard deviation $s_{\hat{\beta}_2} = 0.2256$. The null hypothesis is $\beta_2 \leq 0.3$. Under H_0, we take $\beta_2 = 0.3$, so the quantity

$$t = \frac{\hat{\beta}_2 - 0.3}{0.2256}$$

has a Student's t distribution with 19 degrees of freedom. Note that the degrees of freedom for the t statistic is equal to the degrees of freedom for error. The value of the t statistic is $(0.5652 - 0.3)/0.2256 = 1.1755$. The P-value is between 0.10 and 0.25. (Software yields $P = 0.127$.) It is plausible that $\beta_2 \leq 0.3$.

Page 624

12

Example

5. Find a 95% confidence interval for the mean response μ_{y_i}, and a 95% prediction interval for a future observation when Speed=25 and Pause=15.

95% CI	95% PI
(71.812, 76.732)	(67.641, 80.903)

$$T = \frac{\hat{Y} - (\beta_0 + \beta_1 x_0)}{\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}}$$

$$T = \frac{\hat{Y} - (\beta_0 + \beta_1 x_0)}{\hat{\sigma}\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}}$$

A level $100(1-\alpha)\%$ confidence interval for the quantity $\beta_0 + \beta_1 x$ is given by

$$\hat{\beta}_0 + \hat{\beta}_1 x \pm t_{n-2,\alpha/2} \cdot s_{\hat{y}}$$

where $s_{\hat{y}} = s\sqrt{\frac{1}{n} + \frac{(x-\bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$.

A level $100(1-\alpha)\%$ prediction interval for the quantity $\beta_0 + \beta_1 x$ is given by

$$\hat{\beta}_0 + \hat{\beta}_1 x \pm t_{n-2,\alpha/2} \cdot s_{\text{pred}}$$

where $s_{\text{pred}} = s\sqrt{1 + \frac{1}{n} + \frac{(x-\bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$.

13

Verifying the assumptions: Diagnostic Plots

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \ldots + \beta_k X_{ki} + \varepsilon_i$$
$$\varepsilon_i \sim N(0, \sigma^2)$$

		Assumptions on ε_i:
1) e_i^* (or e_i) versus x_i	$\longleftrightarrow$	1) ε_i's (so $E(\varepsilon_i)$ and $V(\varepsilon_i)$) do not depend on x_i.
2) e_i^* (or e_i) versus y_i	$\longleftrightarrow$	2) ε_i's (so $E(\varepsilon_i)$ and $V(\varepsilon_i)$) do not depend on y_i.
3) normal probability plot of e_i^* (or e_i)	$\longleftrightarrow$	3) ε_i's are normal.
4) e_i^* (or e_i) versus i	$\longleftrightarrow$	4) ε_i's are independent over time. i.e., do not depend on i.

14

Verifying the assumptions: Diagnostic Plots

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \ldots + \beta_k X_{ki} + \varepsilon_i$$
$$\varepsilon_i \sim N(0, \sigma^2)$$

5) $\hat{y}_i$ versus y_i ⟷ 5) close to a 45° line??

6) y_i versus x_i ⟷ 6) an observation with large influence

15

Example:

$$r^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

A Final Exam Question

Regression Analysis: LR versus EI

The regression equation is
LR = 0.622 + 0.0392 EI

Predictor	Coef	SE Coef	T	P
Constant	0.62206	0.02445	25.44	0.000
EI	0.039221	0.007856	4.99	0.000

S = 0.0885719 R-Sq = 24.2% R-Sq(adj) = 23.2%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	0.19551	0.19551	24.92	0.000
Residual Error	78	0.61191	0.00784		
Total	79	0.80742			

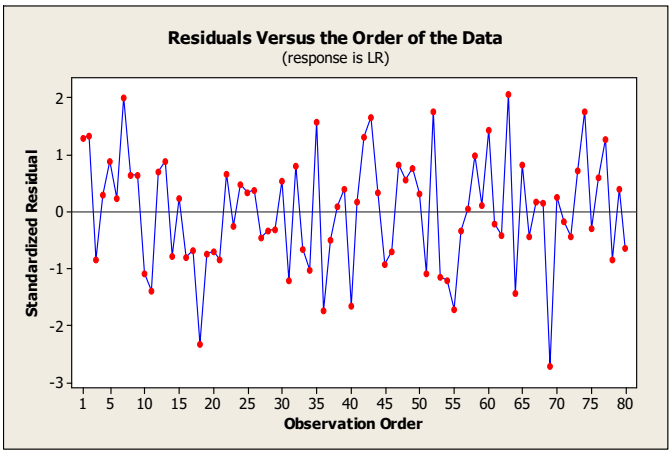
16

Linear regression model

- LR equation: $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ $\quad\quad$ $\varepsilon_i \sim N(0, \sigma^2)$
- Assumptions:
  - Random errors are normal.
  - Random errors are independent of x and y values as well as the order i.
  - Variance of errors is constant.
  - There exists a linear relationship between x and y.
- Validate the assumptions!

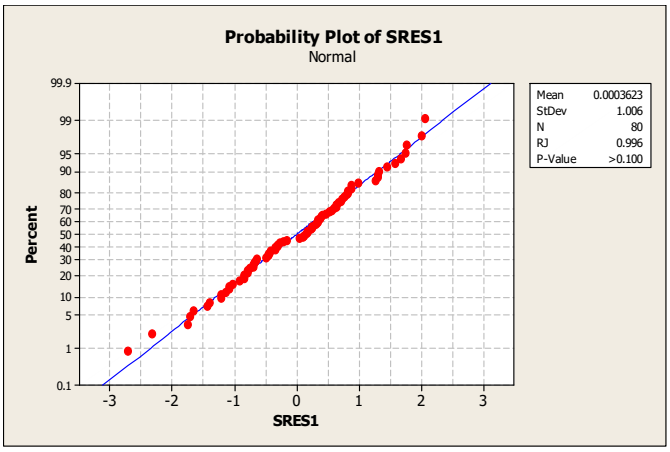
17

Example:
A Final Exam Question



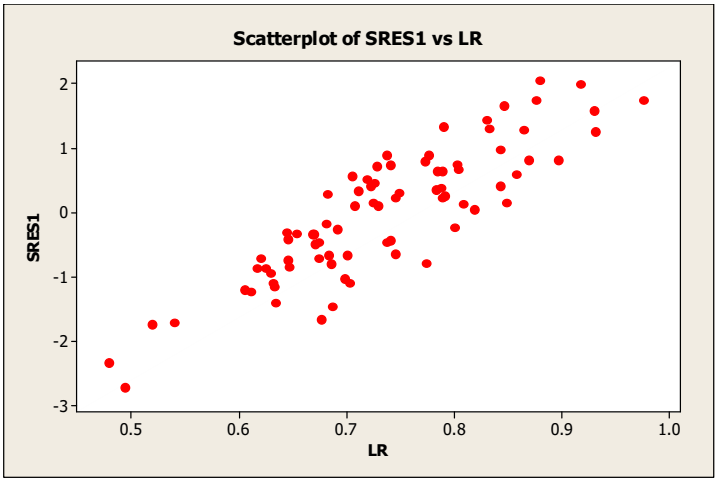
18

Example:
A Final Exam Question



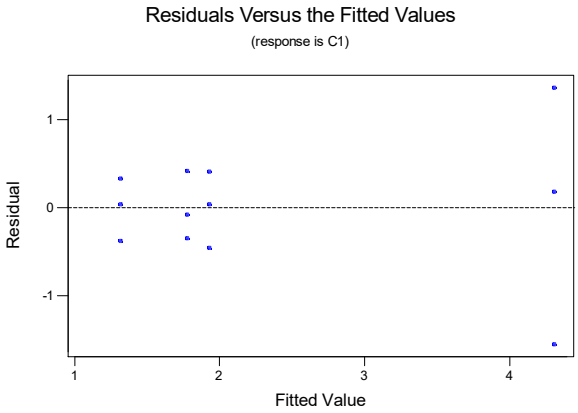
19

Example:
A Final Exam Question



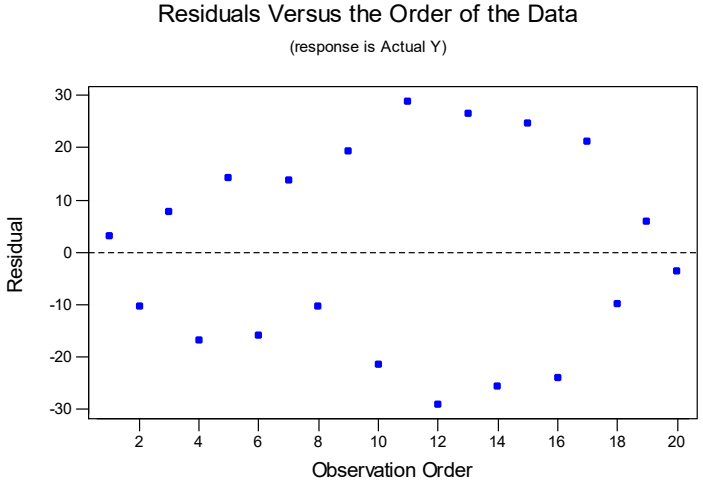
20

A change in the variance of errors with respect to the fitted values



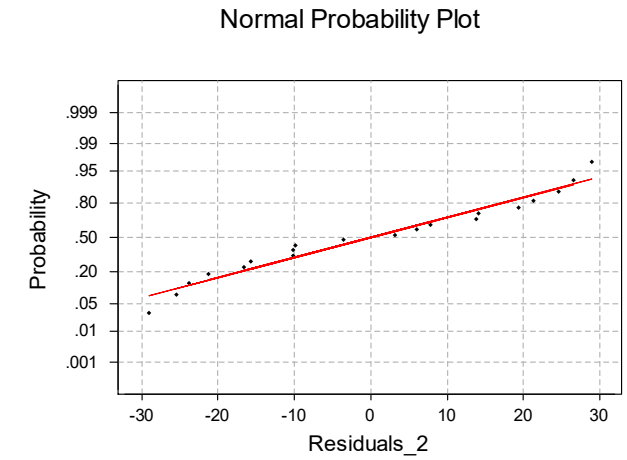
21

Autocorrelation



22

Autocorrelation



3

23

More examples on

- Validating the assumptions of linear regression model

24

Example

- 4 data sets:

Data set	1-3	1	2	3	4	4
Variable	x	y	y	y	x	y
	10.00	8.04	9.14	7.46	8.00	6.58
	8.00	6.95	8.14	6.77	8.00	5.76
	13.00	7.58	8.74	12.74	8.00	7.71
	9.00	8.81	8.77	7.11	8.00	8.84
	11.00	8.33	9.26	7.81	8.00	8.47
	14.00	9.96	8.10	8.84	8.00	7.04
	6.00	7.24	6.13	6.08	8.00	5.25
	4.00	4.26	3.10	5.39	19.00	12.50
	12.00	10.84	9.13	8.15	8.00	5.56
	7.00	4.82	7.26	6.42	8.00	7.91
	5.00	5.68	4.74	5.73	8.00	6.89

- The summary statistics for all sets are identical, hence the regression line, F and t statistics will all the same, but…

25

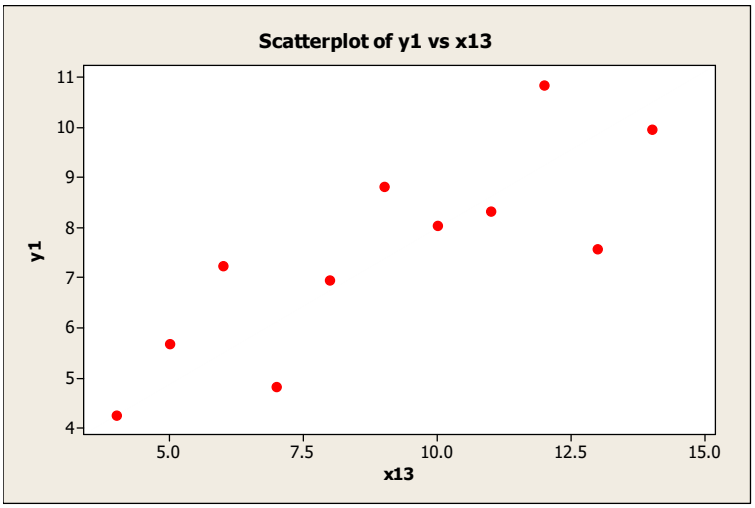
Example: Set 1

The regression equation is: y1 = 3.00 + 0.500 x13

Predictor	Coef	SE Coef	T	P	
Constant	3.00	1.125	2.67	0.026	
x	0.50	0.1179	4.24	0.002	
S = 1.2366		R-Sq = 66.7%		R-sq-adj 62.9%	
ANOVA					
Source	df	SS	MS	F	P
Regression	1	27.510 SSR	27.510 MSR	17.99	0.002
Error	9	13.763 SSE	1.529 MSE		
Total	10	41.273 SST			

26

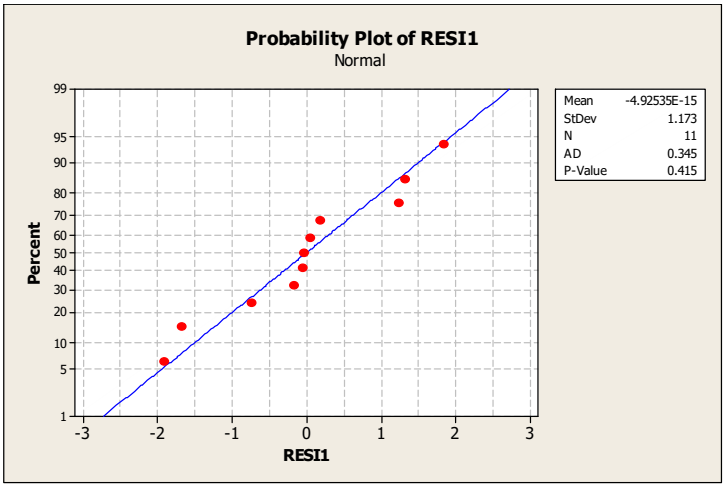
Example: Set 1



27

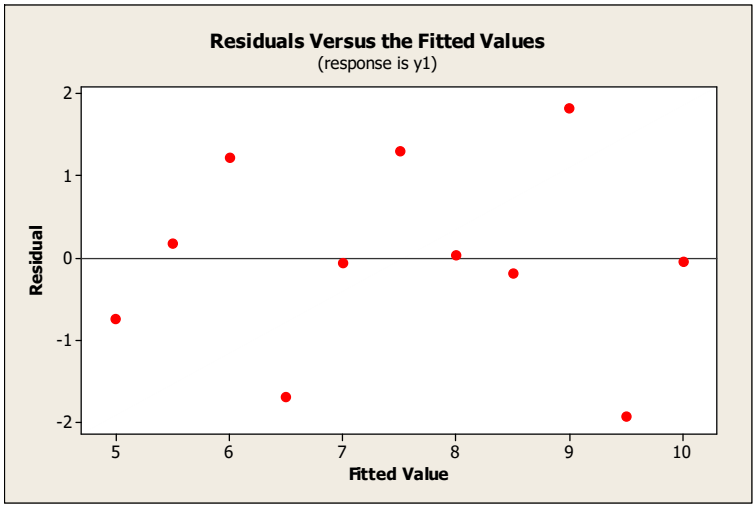
27

Example: Set 1



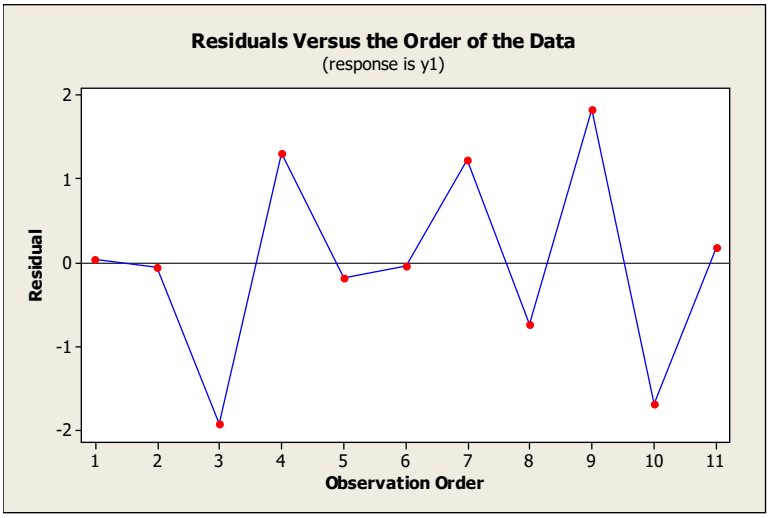
28

Example: Set 1



29

Example: Set 1



30

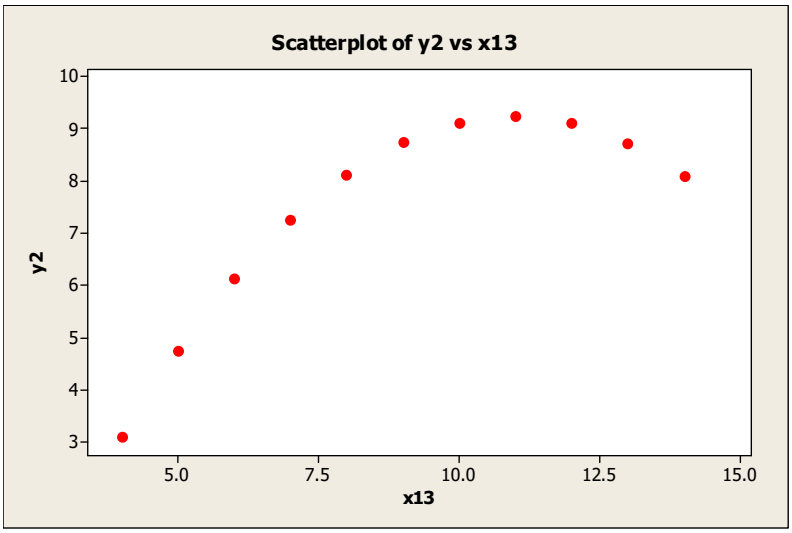
Example: Set 2

The regression equation is: y2 = 3.00 + 0.500 x13

Predictor	Coef	SE Coef	T	P	
Constant	3.00	1.125	2.67	0.026	
x	0.50	0.1179	4.24	0.002	
S = 1.2366		R-Sq = 66.7%		R-sq-adj 62.9%	
ANOVA					
Source	df	SS	MS	F	P
Regression	1	27.510	27.510	17.99	0.002
Error	9	13.763	1.529		
Total	10	41.273			

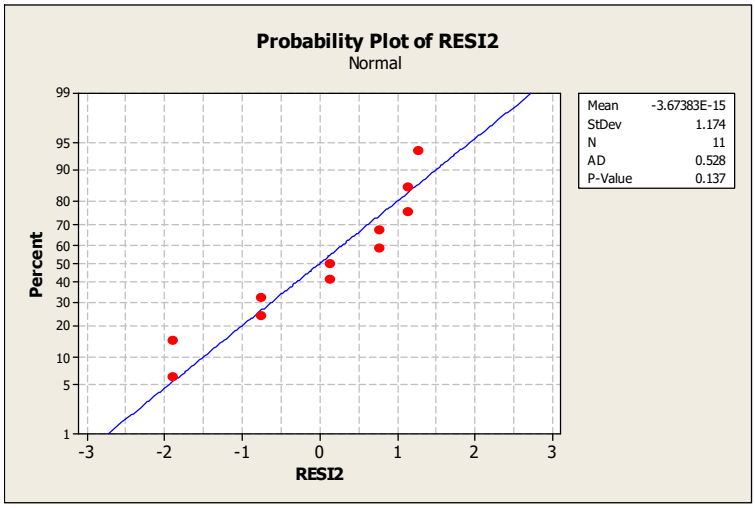
31

Example: Set 2



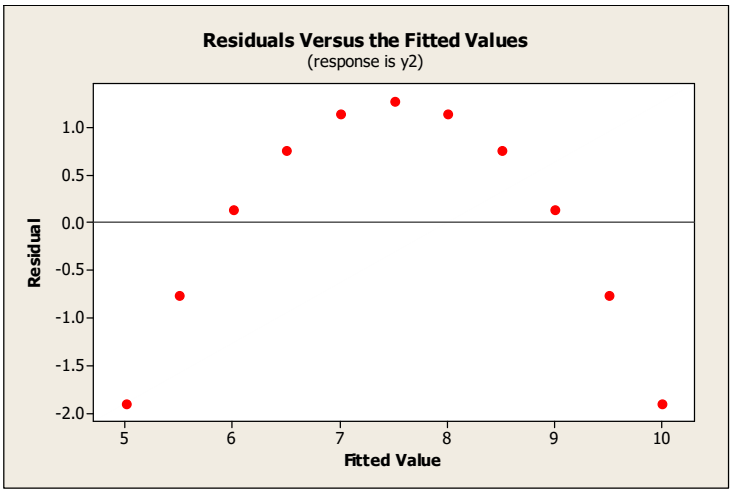
32

Example: Set 2



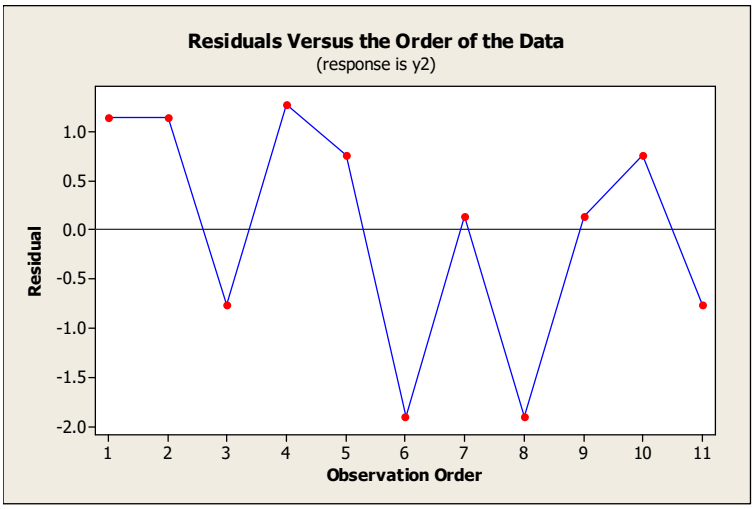
33

Example: Set 2



34

Example: Set 2



35

Example: Set 3
with an influential observation

y = 3.00 + 0.500 x

Predictor	Coef	SE Coef	T	P
Constant	3.002	1.124	2.67	0.026
x	0.4997	0.1179	4.24	0.002

S = 1.236 R-Sq = 66.6% R-Sq(adj) = 62.9%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	27.470	27.470	17.97	0.002
Residual Error	9	13.756	1.528		
Total	10	41.226			

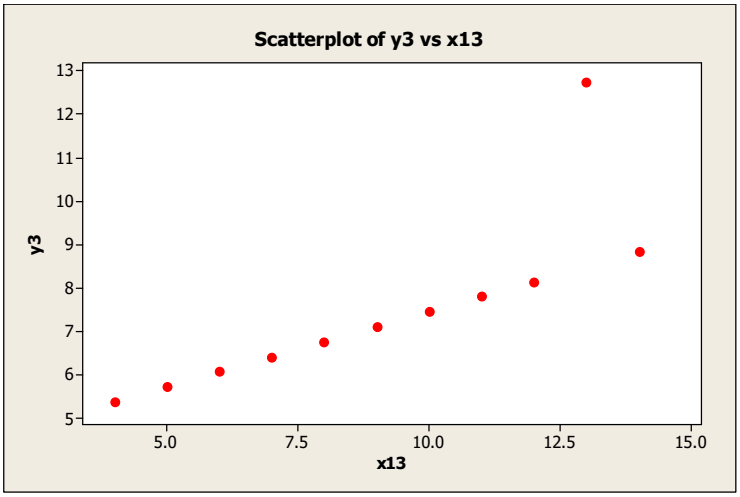
Unusual Observations

Obs	x	y	Fit	SE Fit	Residual	St.R.
3	13.0	12.740	9.499	0.601	3.241	3.00R

R denotes an observation with a large standardized residual

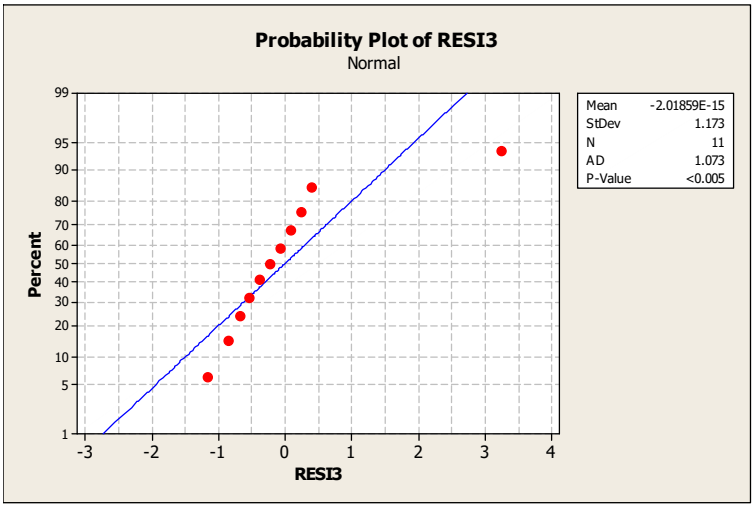
36

Example: Set 3



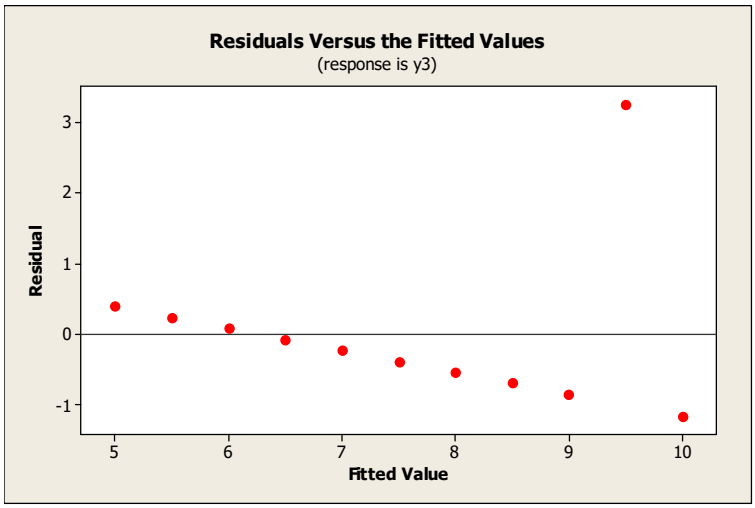
37

Example: Set 3



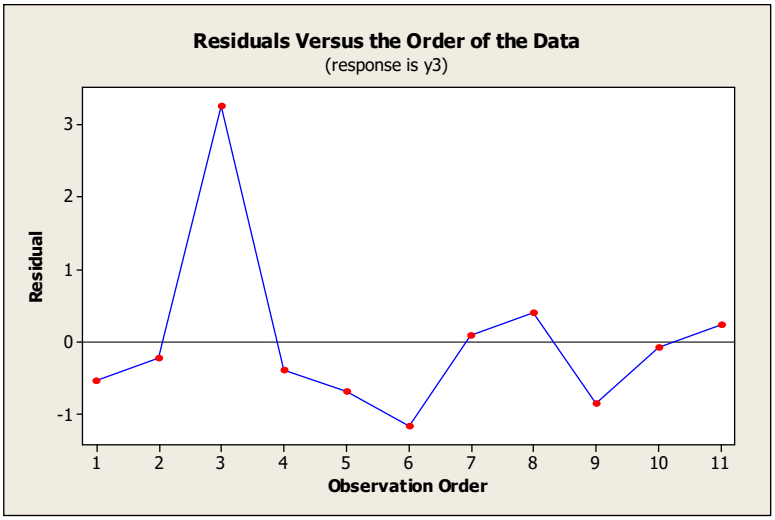
38

Example (set 3)
with an influential observation



39

Example: Set 3
with an influential observation



40

Example: Set 3
without the influential observation

y′ = 4.01 + 0.345 x′

Predictor	Coef	SE Coef	T	P
Constant	4.00565	0.00292	1369.81	0.000
x′	0.345390	0.000321	1077.35	0.000

S = 0.003082 R-Sq = 100.0% R-Sq(adj) = 100.0%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	11.023	11.023	1.161E+06	0.000
Residual Error	8	0.000	0.000		
Total	9	11.023			

41

Example (set 4)
with an influential observation

y = 3.00 + 0.500 x

Predictor	Coef	SE Coef	T	P
Constant	3.002	1.124	2.67	0.026
x	0.4999	0.1178	4.24	0.002

S = 1.236 R-Sq = 66.7% R-Sq(adj) = 63.0%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	27.490	27.490	18.00	0.002
Residual Error	9	13.742	1.527		
Total	10	41.232			

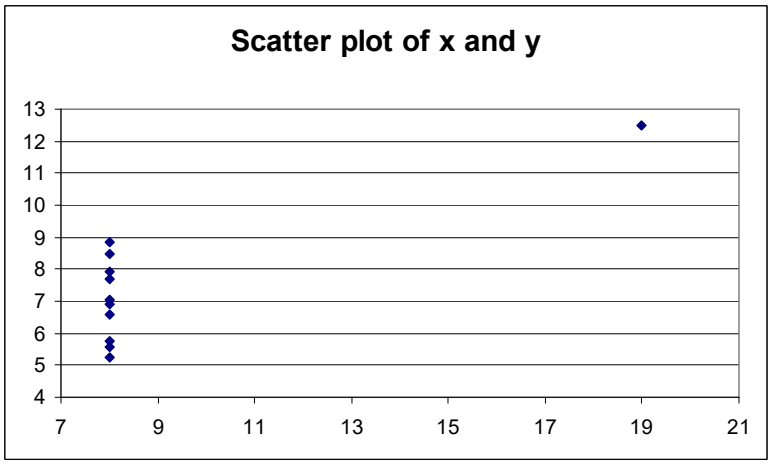
Unusual Observations

Obs	x	y	Fit	SE Fit	Res.	St.R.
8	19.0	12.500	12.500	1.236	0.0	* X

X denotes an observation whose X value gives it large influence.

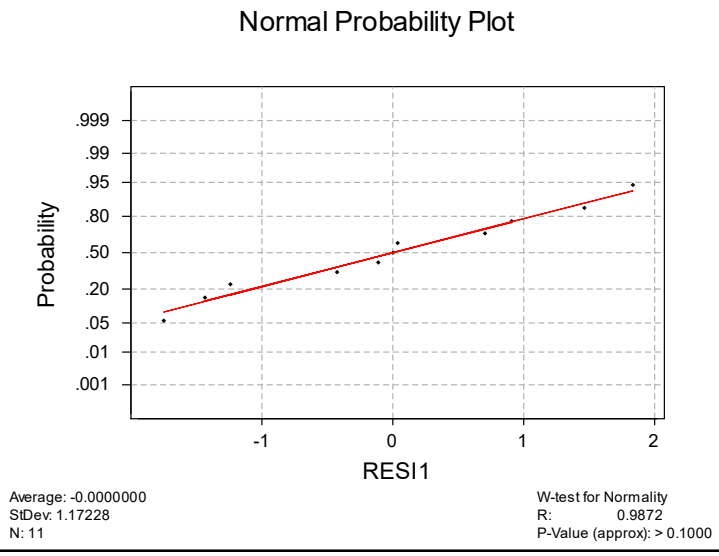
42

Example (set 4)
with an influential observation



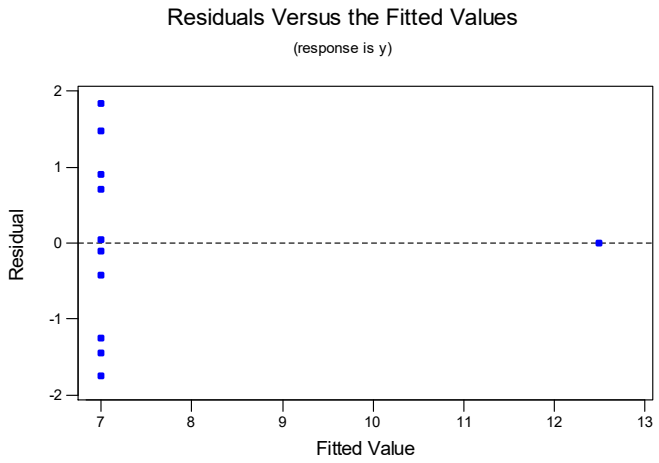
43

Example (set 4)
with an influential observation



44

Example (set 4)
with an influential observation



5

45

Example (set 4)
with an influential observation

- What if we remove the influential observation from the data set?

46

Confounding and Collinearity

- *Fitting separate models to each variable is not the same as fitting the multivariate model.*
- An example is an ideal way to see that confounding and collinearity are something that we need always consider and that justify the use of multiple regression models.

47

Example

A total of 255 gas wells received hydraulic fracturing in order to increase production. In this treatment, a mixture of fluid and sand is pumped into the well. The sand holds open the cracks in the rock, thus increasing the flow of gas.

The main questions are these: Does increasing the volume of fluid pumped increase the production of the well? Does increasing the volume of sand increase the production of the well?

48

Example

Other things being equal, deeper wells produce more gas, because they provide more surface through which the gas can permeate.

For this reason, it is appropriate to express all variables in units per foot of depth of the well. Thus production is measured in units of ft^3 of gas per ft of depth, fluid is measured in units of gal/ft, and sand is measured in units of lb/ft.

Both fluid and sand appear to be strongly related to production.

49

Example

We can use sand to predict production or fluid to predict production. If we fit a simple model, then sand and fluid in their models show up as important predictors.

We might be tempted to conclude that increasing the volume of fluid or the volume of sand would increase production.

There is confounding in this situation. If we increase the volume of fluid, then we also increase the volume of sand.

If production depends only on the volume of sand, there will still be a relationship in the data between production and fluid, and vice versus.

50

Simple Linear Regression Models

The following output presents results using only one independent variable (fluid or sand) in the model. Note that log transformations have been done. Both fluid and sand have a statistically significant effect.

The regression equation is

In Prod = −0.444 + 0.798 In Fluid

Predictor	Coef	SE Coef	T	P
Constant	−0.4442	0.5853	−0.76	0.449
In Fluid	0.79833	0.08010	9.97	0.000

$S = 0.7459$ $R-Sq = 28.2\%$ $R-Sq(adj) = 27.9\%$

The regression equation is

In Prod = −0.778 + 0.748 In Sand

Predictor	Coef	SE Coef	T	P
Constant	−0.7784	0.6912	−1.13	0.261
In Sand	0.74751	0.08381	8.92	0.000

$S = 0.7678$ $R-Sq = 23.9\%$ $R-Sq(adj) = 23.6\%$

51

Multiple Regression

This output presents results from multiple linear regression, in which both fluid and sand are included in the model. In contrast to the separate simple linear regression results, only fluid has a statistically significant effect; sand does not.

The regression equation is

In Prod = −0.729 + 0.670 In Fluid + 0.148 In Sand

Predictor	Coef	SE Coef	T	P
Constant	−0.7288	0.6791	−1.08	0.279
In Fluid	0.6701	0.1687	3.97	0.000
In Sand	0.1481	0.1714	0.86	0.389

$S = 0.7463$ $R-Sq = 28.4\%$ $R-Sq(adj) = 27.8\%$

52

Multiple Linear Regression

- The following regression equation comes from the multiple linear regression output.

  Production = –0.729 + 0.670 Fluid + 0.148 Sand

- The equation of the line uses both variables.

53

Solution

- Multiple regression provides a way to resolve the issue of confounding.
- Here we fit a model with sand and fluid. From this, we can determine which has an effect on production.
- Whether we fit this multiple model or a simple model the R^2 is not particularly high in any case.
- This indicates that there are other important factors affecting production that have not been included in the models.

54

Collinearity

- When two independent variables are very strongly correlated, multiple regression may not be able to determine which is the important one.
- In this case, the variables are said to be **collinear**. The word *collinear* means to lie on the same line, and when two variables are highly correlate, their scatterplot is approximately a straight line.
- The word *multicollinear* is sometimes used as well, meaning that multiple variables are highly correlated with each other.
- When collinearity is present, the set of independent variables is sometimes said to be **ill-conditioned**.

55

Comments

- Sometimes two variables are so correlated that multiple regression cannot determine which is responsible for the linear relationship with *y*.
- In general, there is not much that can be done when variables are collinear.
- The only way to fix the situation is to collect more data, including some values for the independent variables that are not on a straight line.

56

Model Selection

- There are many situations in which a large number of independent variables have been measured, and we need to decide which of them to include in a model.
- This is the problem of **model selection**, and it is a difficult one.
- In practice, model selection often proceeds by ad hoc methods, guided by whatever physical intuition may be available.
- We will not attempt a complete discussion of this extensive and difficult topic. Instead, we will be content to state some basic principles and to present some examples.

57

Model Selection and Occam's Razor

- Good model selection rests on this basic principle known as Occam's razor:
  "The best scientific model is the simplest model that explains the observed data."
- In terms of linear models, Occam's razor implies the **Principle of Parsimony**:
  "A model should contain the smallest number of variables necessary to fit the data."

58

Some Exceptions

1. A linear model should always contain an intercept, unless physical theory dictates otherwise.
2. If a power X^n of a variable is included in the model, all lower powers $X, X^2, \ldots, X^{n-1}$ should be included as well, unless physical theory dictates otherwise.
3. If a product $x_i x_j$ of two variables is included in a model, then the variables x_i and x_j should be included separately as well, unless physical theory dictates otherwise.

59

Notes

- Models that include only the variables needed to fit the data are called **parsimonious** models. Much of the practical work of multiple regression involves the development of parsimonious models.
- Adding a new variable to a model can substantially change the coefficients of the variables already in the model.

60

Can a Variable Be Dropped?

- It often happens that one has formed a model that contains a large number of independent variables, and one wishes to determine whether a given subset of them may be dropped from the model without significantly reducing the accuracy of the model.
- Assume that we know that the model

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \beta_{k+1} x_{k+1i} + \cdots + \beta_p x_{pi} + \varepsilon_i$$

is correct. We will call this the **"full" model**.

61

Null Hypothesis

- We wish to test the null hypothesis

$$H_0: \beta_{k+1} = \cdots = \beta_p = 0$$

- If H_0 is true, the model will remain correct if we drop the variables $X_{k+1}, \ldots, X_p$, so we can replace the full model with the following **reduced model**:

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \varepsilon_i$$

62